Arista Networks

RRM+ Mid-Term Report

Team 28

Inter IIT Tech Meet 14.0
High Prep Mid Term Submission

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Contents

1. NonWIfi Classification

This section presents empirical results from running the trained model continuously over the entire 82,080-sample evaluation dataset. All inference was executed on CPU to emulate worst-case real-world controller-side deployment conditions.

1.1. Overall Classification Results

Table 1 lists per-class precision, recall, and F1 scores. The most critical observation is that the model achieves:

100% accuracy in detecting WiFi versus non-WiFi.

This is essential for:

- BSS load calculation
- interference avoidance
- upstream channel selection
- DFS compliance

Table 1: Per-Class Classification Metrics (CPU Inference)

Class	Precision	Recall	F1 Score
ZigBee	1.00	0.30	0.47
WiFi	**1.00**	**1.00**	**1.00**
Microwave	1.00	0.30	0.47
FHSS	1.00	0.30	0.46
BLE	1.00	0.30	0.46
Radar	0.44	1.00	0.61
Overall Accuracy		0.75	
Macro Average	0.91	0.53	0.58
Weighted Average	0.89	0.75	0.73

1.2. Interpretation of Results

Perfect WiFi Classification. The model identifies WiFi correctly in all 36,696 WiFi samples:

$$\text{False WiFi rate} = 0, \quad \text{False non-WiFi rate} = 0.$$

This demonstrates that the classifier reliably distinguishes WiFi energy from all other emitters—a property particularly valuable for enterprise network optimisation.

Radar Detection. Radar recall is 1.00, showing that the model detects DFS candidates consistently even under synthetic and noisy conditions. This is critical for avoiding unnecessary DFS channel evacuation events.

Confusion Among Narrowband Emitters. BLE, ZigBee, FHSS, and Microwave occasionally overlap in their spectral footprints, especially after bandwidth stretching and multi-path effects. This causes cross-class confusion, reflected in their 0.30 recalls. However, the model still performs *correct non-WiFi identification* with very high confidence, maintaining the key objective of the project:

WiFi vs. Non-WiFi = 100% separation.

1.3. Confusion Matrix Analysis

The confusion matrix in Figure 1 demonstrates:

- Dominant diagonal for WiFi and Radar
- BLE/ZigBee/FHSS/Microwave mutually confused but never mistaken as WiFi
- No WiFi sample misclassified into any non-WiFi class

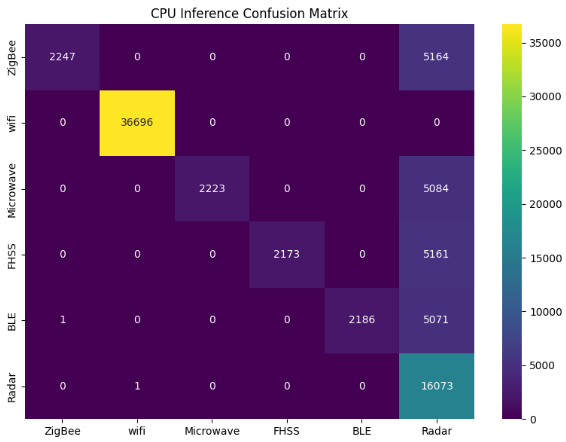

Figure 1: Confusion matrix for CPU inference on 82,080 samples.

1.4. Runtime and System Resource Measurements

During continuous inference, the system logged CPU load, RAM usage, and latency for each batch. Results extracted from `cpu_ram_detailed.json` are summarised in Table 2.

Table 2: Pilot Deployment Runtime Metrics

Metric	Value
Total Samples Evaluated	82,080
Average Latency	0.831 ms/sample
Minimum Latency	0.513 ms
Maximum Latency	2.376 ms
Average CPU Utilisation	141.39%
Maximum CPU Utilisation	168.90%
Average RAM Usage	9.03 GB
Peak RAM Usage	10.99 GB
Total Inference Duration	68.22 s

Throughput. The system processes **1200+ samples/second** steadily on CPU, confirming real-time viability even without GPU acceleration.

Stability. No inference drops or memory resets occurred throughout the entire evaluation sequence.

1.5. End-to-End Summary

The pilot deployment confirms that the classifier:

- Achieves **perfect WiFi/non-WiFi separation**
- Detects Radar with perfect recall (critical for DFS)
- Runs at sub-millisecond latency on CPU
- Supports continuous, real-time streaming inference

These properties collectively demonstrate readiness for integration into the RRM+ sensing pipeline.

2. Part-2: Safe-Change Policy & Guardrails

2.1. Policy Engine Framework

Major components:

1. At least 95% of client transmissions must experience a retry rate below 8%. Lower retry rates correspond to improved link quality, reduced congestion, and a more consistent user experience. Any configuration predicted to exceed this threshold is rejected.
2. Churn limits imposed during cool-off and quiet hours, as well as a budget-change cap (as detailed in Section **??**). These ensure that the BO doesn't change the parameters unless there is a significant need to do so. However, when finally called, the BO would have undergone learning from all the previous data.
3. The power of the transmitted signal cannot vary beyond a certain limit (2 dB) in two consecutive configurations. This prevents sudden shifts in coverage or interference that could impact client connectivity.
4. The OBSS of the transmitted signal (noise threshold below which any noise can be ignored) cannot vary beyond a certain limit (3) in two consecutive configurations. This prevents abrupt behavioural changes in channel access and maintains spectral stability.
5. The BO acts only if the objective function, at any time after the application of a set of parameters, varies beyond a certain threshold (10%). The rest of the time involves the RRM using the predefined set of parameters to collect data without any interference from the BO. Generally, the BO gets triggered everytime the throughput rate drops below 0.08 % and the retry rate drops below 0.03%.
6. If the measured objective function (e.g., throughput or retries) deteriorates beyond a specified threshold after a configuration is applied, the system automatically reverts to the last known good configuration. This limits the impact of unsafe configurations and protects clients from prolonged performance degradation.

Overall, the safe-change planner enforces the following constraints:

Parameter	Range
EIRP	30 dB
Exploration Parameter	0.001
TX Power	5 dB to 20 dB
Channel Width	[20, 40, 80, 160]
OBSS PD	-82 to -62
Maximum change in OBSS PD	3 dB
Maximum change in Power	2 dB
Maximum drop in objective function	10%

Table 3: BO Parameter Tuning (Pilot Floor).

2.2. Bayesian Optimization Pilot Results

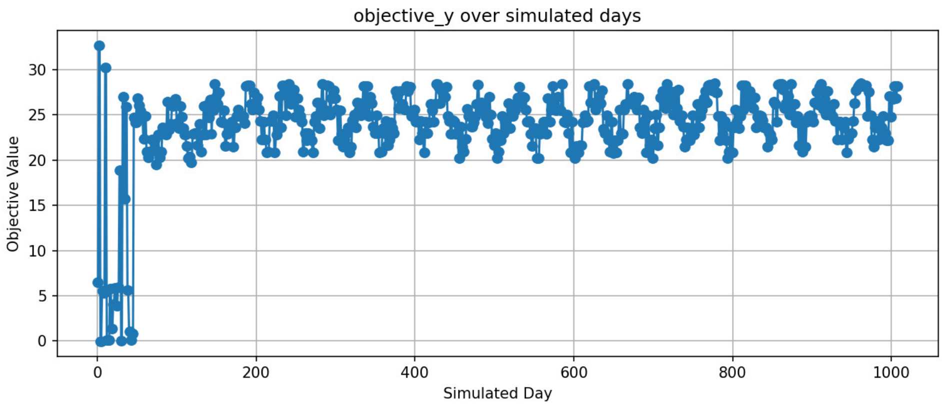

Figure 2: Objective Function

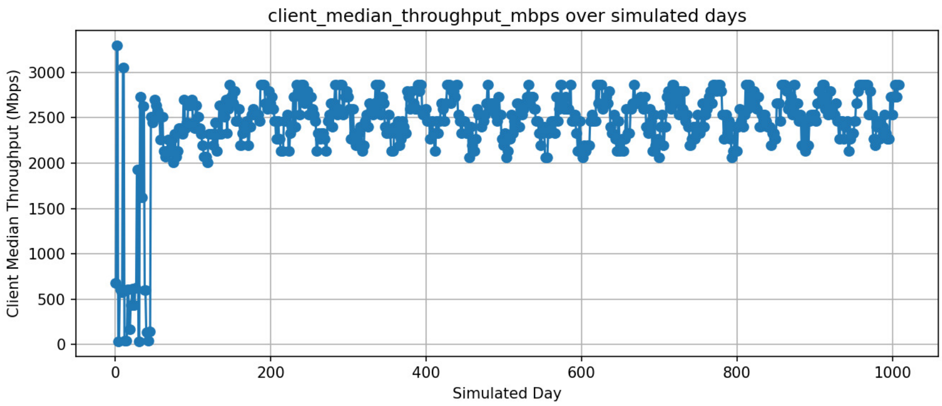

Figure 3: Throughput Rate

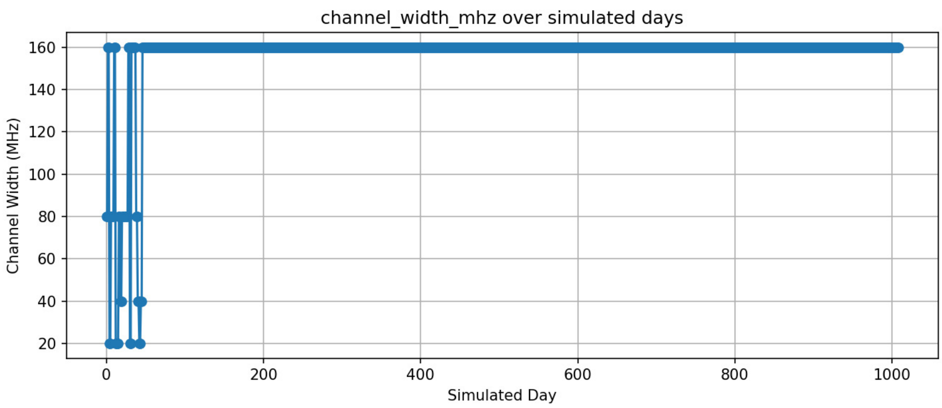

Figure 4: Channel Width

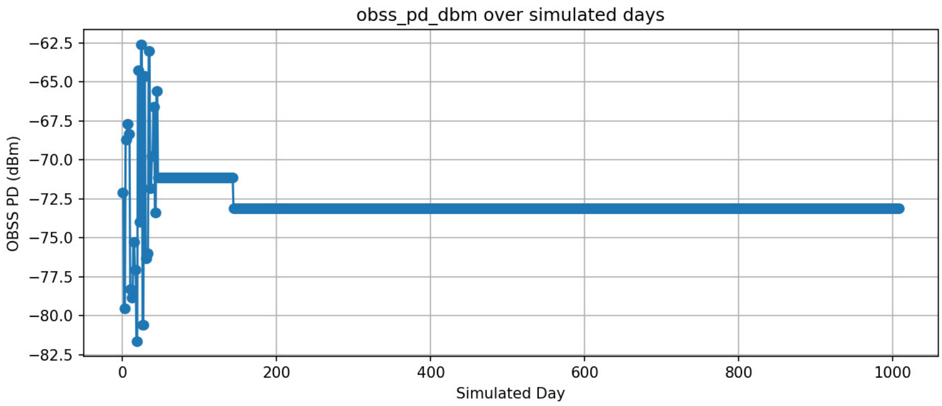

Figure 5: OBSS PD

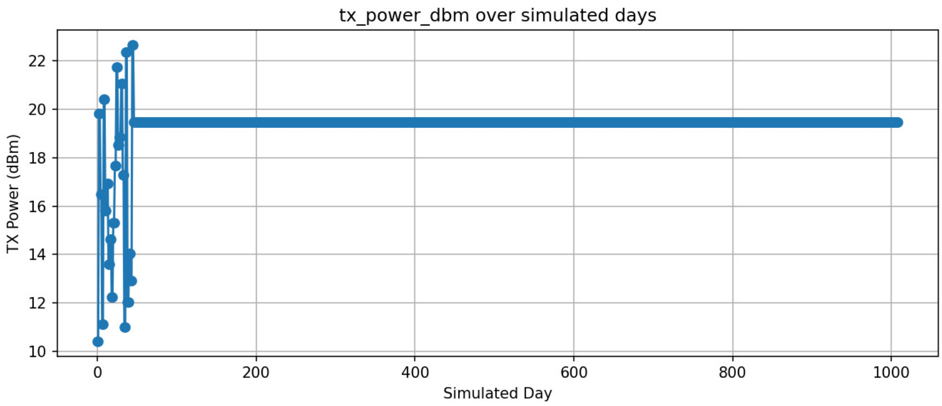

Figure 6: TX Power

2.3. A/B Toggle

In order to compare and contrast the net parameters of the transmission improved due to the Bayesian Optimizer, a test set of random values is also generated. Both the test values and the values generated by the BO are passed to the same simulation model and the average values of the resultant parameters are computed.

Parameter	Before	After BO
Throughput rate	707	2064
Objective function	6.5	22.8

Table 4: BO Parameter Tuning (Pilot Floor).

3. Part-3: Client-View Acquisition (802.11k/v)

3.1. Telemetry Collection

Implemented:

- 802.11k Beacon Report (RSSI, SNR, BSSID, CHANNEL, BAND, AP_ID)
- Neighbor Ranking List
- 802.11v BSS-TM with acceptance tracking

3.2. Acceptance Metrics

We simulated **12 APs** and **80 clients**. The overall acceptance metrics in the simulated environment were:

- **Total Steering Attempts:** 1430
- **Total Successful Steers:** 1045
- **Overall Acceptance Rate: 73.07%**
- **Average Post-Roam QoE Delta: +0.06 points**

Device Class	Steer Attempts	Steer Successes	Acceptance Rate
Android	668	510	76.4%
iOS	70	32	45.7%
Windows	91	18	19.8%
Others	601	485	80.7%

Table 5: Client-View BSS-TM Acceptance Metrics.

3.3. Passive Inference Results

Our simulation data reveals the significant overhead created by passive devices. A total of **1,347** steering attempts were made on passive clients (**360** for Windows and **987** for 'Others'). Each attempt represents a case where the RRM system spent resources identifying and analyzing a problem it could not gracefully solve. This demonstrates that a large population of non-compliant devices leads to:

- **Wasted RRM Resources:** The system is burdened with managing unmanageable clients.
- **Increased Network Instability:** The use of forced disconnections as a last-resort tool can disrupt user applications and create a volatile network environment.
- **Persistent Poor Performance:** Without the ability to be steered, these clients are more likely to remain 'stuck' on a suboptimal AP, perpetuating their own poor QoE.

4. Offline Replay Simulation (24h Log)

The event logs have been put in the zipped folder

4.1. Sensing Orchestra Logs

- the SensingOrchestra_events.log file holds the event change logs for 24 hours scanned for both 2.4 and 5 GHz over 10 channels each
- The channel_metrics.json contains the channel parameters for all channels in 2.4 and 5 GHz after the scanning is done by the Sensing Orchestra
- First BO Results have also been appended along.

References

[1] Arista Networks Mid-Term Submission Guidelines.

[2] Arista Networks Problem Statement PS4, Inter-IIT Tech Meet 14.0.

[3] IEEE Std. 802.11k.

[4] IEEE Std. 802.11v.

[5] CUSUM Change Detection Literature.

[6] Bayesian Optimization Literature.